```
    deepspeed="ds_config.json"  # DeepSpeed config file
)
```

Optimising training

Using FP16 precision (instead of FP32) speeds up training and reduces memory consumption.

```
training_args = TrainingArguments(fp16=True, output_dir="./results")
```

Checkpointing

Saving checkpoints ensures training can resume after interruptions.

```
training_args = TrainingArguments(save_steps=500, save_total_limit=3, output_dir="./results")
```

Evaluation and deployment of machine learning models

Evaluating a machine learning model is a crucial step to ensure its performance and reliability before deployment. Various metrics and benchmark datasets are used to assess the effectiveness of the model.

Evaluation metrics: Depending on the type of model and task, different evaluation metrics are used.

- *Perplexity:* Commonly used in language models, Perplexity measures how well a probability distribution predicts a sample.
- *BLEU (Bilingual Evaluation Understudy):* Evaluates the quality of machine-generated translations against human references.
- *ROUGE (Recall-Oriented Understudy for Gisting Evaluation):* Measures the overlap of n-grams between the model's output and a reference text, often used for text summarisation.
- *Accuracy, Precision, Recall, F1-score:* Used in classification models to measure correctness and performance.
- *Mean Squared Error (MSE) and R-squared:* Applied in regression tasks to measure prediction errors.

Benchmark datasets: Benchmark datasets provide standardised comparisons for model evaluation (Table 5).

GLUE	A collection of tasks for evaluating natural language understanding (NLU) models.
SuperGlue	An advanced version of GLUE designed for more complex language tasks
ImageNet	A widely used dataset for image classification models.
MS COCO	Used for object detection and image captioning models.
SQuAD	A dataset for evaluating question-answering models.

Table 5: Benchmark datasets

Once a model is evaluated and optimised, it is deployed to serve real-world applications. Several tools and frameworks facilitate deployment (Table 6).

ONNX	Enables interoperability between different deep learning frameworks, allowing models trained in one framework to be deployed in another.
TensorRT	A high-performance deep learning inference library optimised for NVIDIA GPUs.
Hugging Face Inference API	Provides pre-trained models with an easy-to-use API for quick deployment.

Table 6: Deployment tools

Building your own LLM may seem daunting, but with the right open source tools, the process is more accessible than ever. The key is to start small—experiment with fine-tuning an existing model before scaling up to full-fledged training.

Whether you're building a proprietary AI assistant, a specialised content generator, or a research-driven model, the open source community has provided everything you need.

Ready to build your LLM? Choose your framework, set up your environment, and start training today! The future of AI is open source—be part of the revolution! END

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